Notes: Kacperczyk and Seru (JF, 2007)

Fund Manager Use of Public Information: New Evidence on Managerial Skills

Contents

1 Big picture

- **Question.** Can we learn about mutual fund manager skill by observing how strongly managers change portfolio holdings in response to public information?
- **Core idea.** A skilled manager has more precise private information. Therefore, when public information changes, the skilled manager should adjust holdings less than an uninformed or low-skill manager who relies heavily on public signals.
- **Main measure.** The paper constructs **Reliance on Public Information** (RPI) as the R^2 from a fund-quarter cross-sectional regression of changes in portfolio holdings on changes in past analyst recommendations.
- **Interpretation of RPI.** High RPI means that a large share of a fund's holding changes can be explained by public analyst-recommendation changes. Low RPI means that holding changes are less tied to those public signals and are more consistent with private information or idiosyncratic investment judgment.
- **Theory.** A Grossman–Stiglitz-style noisy rational expectations model shows that the sensitivity of informed investors' holdings to public information falls as the precision of their private signal rises.
- **Prediction 1.** If low RPI captures skill, low-RPI funds should subsequently outperform high-RPI funds.
- **Prediction 2.** If investors recognize that low RPI contains information about skill, low-RPI funds should receive higher future fund flows, even after controlling for past performance.
- **Empirical setting.** The sample contains 1,696 actively managed diversified U.S. equity mutual funds from 1993Q1 to 2002Q4.
- **Baseline result.** Low-RPI funds have higher factor-adjusted and holdings-based performance. A one-standard-deviation increase in RPI lowers the Carhart four-factor alpha by about 0.46% per year.
- **Mechanism evidence.** The holding-based results are strongest for stock selection and style selection, not for characteristic timing. This supports the interpretation that low-RPI managers are better stock pickers rather than simply better market timers.
- **Flow result.** Investors allocate more money to low-RPI funds after controlling for past returns and standard fund characteristics.
- **Robustness.** The evidence survives alternative definitions of RPI, alternative public-information sets based on analyst earnings forecasts, fund fixed effects, manager fixed effects, spillover controls across related stocks, turnover controls, and style/size checks.
- **Takeaway.** The paper proposes a holdings-based diagnostic for managerial skill: not just whether a fund performs well, but whether its trades look like they are driven by public information that everyone already observes.

2 Data and measurement (key objects)

2.1 Why this setting is useful

- Mutual funds disclose detailed portfolio holdings, which makes it possible to study actual portfolio decisions rather than only realized returns.
- Sell-side analyst recommendations provide a stock-level, time-varying proxy for public information.
- The mutual fund setting also permits a second test based on investor flows: if RPI reveals skill, outside investors may reward low-RPI funds with inflows.

Interpretation. The paper links three objects that are usually studied separately: public information, portfolio rebalancing, and fund performance.

2.2 Data sources and sample construction

The paper merges four main datasets:

- **CRSP Survivorship-Bias-Free Mutual Fund Database:** fund returns, total net assets, fees, expenses, investment objectives, and other fund characteristics.
- **CDA/Spectrum holdings data:** quarterly or semiannual stock holdings of mutual funds.
- **IBES analyst recommendation data:** stock-level sell-side analyst recommendations.
- **CRSP stock data:** stock prices used to value and identify holdings.

The final sample excludes balanced funds, bond funds, international funds, index funds, sector funds, and duplicate share classes. The resulting sample contains:

$$\text{1,696 actively managed diversified U.S. equity funds, 1993Q1–2002Q4.} \tag{1}$$

The IBES recommendation data contain 8,993 analysts covering 7,766 firms. Recommendations are coded on an inverse scale:

$$1 = \text{strong buy}, \qquad 5 = \text{strong sell}. \tag{2}$$

Therefore, an upgrade corresponds to a negative numerical change.

Interpretation. The key empirical move is to ask whether funds buy or sell stocks in ways that can be statistically explained by lagged public analyst-recommendation changes.

2.3 Main dependent variable in the RPI construction

For stock i in fund m at time t, the dependent variable is the percentage change in split-adjusted holdings:

$$\%\Delta\text{Hold}_{i,m,t}. \tag{3}$$

When a fund initiates a new position, the percentage change would be infinite. The paper bounds such changes at 100%, with robustness checks using other bounds.

Interpretation. The object being explained is not the level of holdings, but the change in holdings. This focuses on the manager's active portfolio decision.

2.4 Public information: analyst recommendation changes

The main public-information variables are lagged changes in consensus analyst recommendations:

$$\Delta \mathrm{Re}_{i,t-p}, \qquad p = 1, 2, 3, 4. \tag{4}$$

The paper uses past recommendation changes rather than contemporaneous changes to reduce concerns that analysts react to fund trades.

Interpretation. The measure is about reliance on already-public information. If public information has already been incorporated into prices, simply following it should not by itself generate abnormal performance.

2.5 The RPI measure

For each fund m and quarter t, the paper estimates a cross-sectional regression across all stocks i in the fund portfolio:

$$\begin{aligned} \%\Delta \mathrm{Hold}_{i,m,t} = \beta_{0,t} &+ \beta_{1,t}\Delta \mathrm{Re}_{i,t-1} + \beta_{2,t}\Delta \mathrm{Re}_{i,t-2} \\ &+ \beta_{3,t}\Delta \mathrm{Re}_{i,t-3} + \beta_{4,t}\Delta \mathrm{Re}_{i,t-4} + \varepsilon_{i,m,t}. \end{aligned} \tag{5}$$

Then RPI is defined as:

$$\mathrm{RPI}_{m,t-1} = 1 - \frac{\sigma^2(\varepsilon_{m,t})}{\sigma^2(\%\Delta \mathrm{Hold}_{m,t})}. \tag{6}$$

Equivalently, RPI is the unadjusted R^2 of the holdings-change regression.

Interpretation. A high R^2 means public analyst information explains much of the fund's trading. A low R^2 means the trades are less explained by public analyst information.

2.6 Descriptive facts about RPI

The paper reports substantial cross-sectional variation:

- mean RPI: 29.0%,
- median RPI: 21.3%,
- standard deviation: 25.2%,
- range: 0.01% to 99.9%.

Interpretation. There is enough dispersion in RPI to rank funds and test whether funds with low reliance on public information perform differently.

2.7 Fund controls and outcome variables

The main control variables are:

- log total net assets, log(TNA),
- log fund age,
- expenses,
- turnover,
- new money growth,
- loads,
- lagged returns,
- total risk where appropriate.

The main performance outcomes are:

- CAPM alpha,
- Fama–French three-factor alpha,
- Carhart four-factor alpha,
- conditional versions of these alphas,
- Daniel–Grinblatt–Titman–Wermers holding-based measures: GT, CS, CT, and AS.

Interpretation. The empirical design checks that RPI predicts performance beyond common risk factors, styles, fund size, age, trading intensity, costs, and flows.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism in the model

The model is a two-period noisy rational expectations economy with:

- one risk-free asset,
- one risky asset with payoff $\tilde{u}$,
- informed investors who observe both private and public signals,
- uninformed investors who observe public signals and learn from price,

- random asset supply that prevents price from fully revealing private information.

The key premise is:

$$\text{managerial skill} \quad \Longleftrightarrow \quad \text{precision of private signal.} \tag{7}$$

Interpretation. The paper treats skill as the ability to acquire or process information that is not already fully reflected in public signals.

3.2 Investor budget and objective

Investor n chooses the risky-asset position x^n. The budget constraint is:

$$m^n + px^n = \bar{m}^n, \tag{8}$$

where m^n is cash, p is the risky-asset price, and $\bar{m}^n$ is initial wealth.

Terminal wealth is:

$$\tilde{w}^n = m^n + \tilde{u}x^n, \tag{9}$$

or equivalently,

$$\tilde{w}^n = \bar{m}^n + (\tilde{u} - p)x^n. \tag{10}$$

With CARA utility and normal distributions, the investor maximizes:

$$\mathbb{E}[\tilde{w}^n] - \frac{\lambda}{2}\operatorname{Var}[\tilde{w}^n]. \tag{11}$$

Interpretation. Demand depends on the investor's posterior mean and posterior precision about the risky asset payoff.

3.3 Signals and price

The informed investor observes:

- private signal s_1,
- public signal s_2.

The public and private signals have precisions ρ_2 and ρ_1, respectively. The equilibrium price has the linear form:

$$p = a\bar{u} + bs_1 + cs_2 - dt + e\bar{t}, \tag{12}$$

where t is random net supply and the coefficients are functions of model primitives.

Interpretation. Price partially reveals the private signal, but random supply prevents full revelation.

3.4 Difference between informed and uninformed holdings

Let x_I be the per capita risky-asset holding of informed investors and x_U the per capita holding of uninformed investors. Define:

$$\Delta \equiv x_I - x_U. \tag{13}$$

The paper derives:

$$\mathbb{E}[x_I - x_U] = \frac{\mathbb{E}[\tilde{u} - p](\rho_1 - \rho_\theta)}{\lambda} > 0, \tag{14}$$

where ρ_θ is the precision of the information inferred by uninformed investors from price.

Interpretation. Informed investors hold more of the risky asset on average because their private signal gives them an informational advantage.

3.5 Key comparative statics

The central derivative is the effect of the public signal on the informed-minus-uninformed holding difference:

$$\frac{\partial \Delta}{\partial s_2} = \frac{\rho_2(\rho_\theta - \rho_1)}{\gamma \lambda} < 0. \tag{15}$$

The sensitivity falls further as the precision of private information increases:

$$\frac{\partial^2 \Delta}{\partial s_2 \partial \rho_1} < 0. \tag{16}$$

Interpretation. Public news moves uninformed investors more than informed investors. More skilled informed investors, with higher ρ_1, rely even less on the public signal.

3.6 Prediction 1: RPI and performance

The main performance regression is:

$$\alpha_{m,t} = \beta_0 + \beta_1 \mathrm{RPI}_{m,t-1} + \gamma \mathrm{Controls}_{m,t-1} + \varepsilon_{m,t}. \tag{17}$$

The paper predicts:

$$\beta_1 < 0. \tag{18}$$

Interpretation. Funds that rely more on public information should perform worse if low reliance on public information is a proxy for private information or skill.

3.7 Performance measures

The paper uses factor-based alphas and holding-based measures.

The Carhart-style return regression uses market, size, value, and momentum factors. The alpha is the abnormal return relative to these passive risk/style factors.

The holding-based characteristic selectivity measure is conceptually:

$$CS_t = \sum_j w_{j,t-1} \left[R_{j,t} - BR_t(j, t-1) \right], \tag{19}$$

where $w_{j,t-1}$ is the portfolio weight in stock j and $BR_t(j, t-1)$ is the benchmark return for a portfolio matched on characteristics.

Interpretation. Factor alphas ask whether the fund beats passive factor benchmarks. Holding-based measures ask whether the actual stocks held beat characteristic-matched benchmarks.

3.8 Prediction 2: RPI and fund flows

The flow regression is:

$$\text{NetFlow}_{m,t} = \beta_0 + \beta_1 \text{RPI}_{m,t-1} + \gamma \text{Controls}_{m,t-1} + \varepsilon_{m,t}. \tag{20}$$

Net flow is measured as:

$$\text{NetFlow}_{m,t} = \frac{\text{TNA}_{m,t} - \text{TNA}_{m,t-1}(1 + R_{m,t})}{\text{TNA}_{m,t-1}}. \tag{21}$$

The prediction is again:

$$\beta_1 < 0. \tag{22}$$

Interpretation. If investors infer skill from low RPI, then low-RPI funds should receive more inflows even after controlling for past returns.

3.9 Alternative RPI measures

The first alternative measure uses the magnitude of the t-statistics on the public-information coefficients:

$$\text{RPI}^{\beta}_{m,t-1} = \sum_{p=1}^{4} \left| \frac{\beta_{p,m,t}}{se(\beta_{p,m,t})} \right|. \tag{23}$$

The second alternative, RPI^{earn}, replaces analyst recommendations with changes in analyst earnings forecasts.

Interpretation. These alternatives check whether the results are driven by the specific R^2 construction or by the specific choice of analyst recommendations as public information.

3.10 Manager fixed effects

To test whether RPI is linked to fund managers rather than only funds, the paper estimates:

$$\text{RPI}_{i,t} = \alpha_t + \gamma_i + \beta X_{i,t} + \lambda_{mgr} + \varepsilon_{i,t}, \tag{24}$$

where α_t are year effects, γ_i are fund fixed effects, $X_{i,t}$ are time-varying fund controls, and λ_{mgr} are manager fixed effects.

Interpretation. If manager effects matter, changes in management should explain meaningful variation in RPI after fund and time effects.

3.11 Spillover-adjusted RPI

Because information about one stock may affect related stocks, the paper also adds industry-level recommendation changes:

$$\%\Delta\text{Hold}_{i,m,t} = \beta_{0,t} + \sum_{p=1}^{4} \beta_{p,t} \Delta\text{Re}_{i,t-p} + \sum_{p=1}^{4} \delta_{p,t} \Delta I\text{Re}_{-i,t-p} + \varepsilon_{i,m,t}, \tag{25}$$

where $\Delta I\text{Re}_{-i,t-p}$ is the recommendation change for other stocks in the same three-digit SIC industry.

Interpretation. This checks whether a fund's holdings react not only to direct public information about stock i, but also to public information about related stocks.

3.12 Risk-taking regressions

The paper studies whether high-RPI funds take more risk:

$$TotRisk_{m,t} = \beta_0 + \beta_1 \text{RPI}_{m,t-1} + \gamma \text{Controls}_{m,t-1} + \varepsilon_{m,t}, \tag{26}$$

and:

$$UnsysRisk_{m,t} = \beta_0 + \beta_1 \text{RPI}_{m,t-1} + \gamma \text{Controls}_{m,t-1} + \varepsilon_{m,t}. \tag{27}$$

Interpretation. If high-RPI managers are less skilled, they may take more risk in an attempt to improve their records.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline relationship

The baseline relationship is identified from variation in lagged RPI across funds and over time, after controlling for standard fund characteristics and quarter time effects.

- The public-information variables are lagged recommendation changes.
- Performance and flows are measured after RPI is constructed.
- The regressions include time fixed effects to absorb market-wide shocks.
- Controls include fund size, age, expenses, turnover, and fund flows.

Interpretation. The empirical design asks whether a fund's recent trading reliance on public information predicts subsequent performance and investor flows.

4.2 Reverse causality

A central concern is that analysts may respond to mutual fund trades rather than funds responding to analysts. The paper reduces this concern by using past analyst recommendation changes rather than contemporaneous changes.

Interpretation. Lagging public information makes the timing more consistent with managers reacting to public signals, not analysts reacting to managers.

4.3 Omitted variables

The paper addresses omitted variables through:

- standard fund controls,
- quarterly time dummies,
- fund fixed effects in robustness tests,
- manager fixed effects in the turnover analysis,
- residualizing RPI with respect to fund characteristics,
- checking size quintiles and investment styles.

Interpretation. The main negative RPI–performance relation is not simply a comparison of large versus small funds, old versus young funds, low-turnover versus high-turnover funds, or one investment style versus another.

4.4 No formal instrument

The paper does not use a conventional instrumental-variable design for RPI. Its credibility comes from a combination of theory, timing, controls, alternative measurement, and robustness tests.

Interpretation. The results are best read as strong evidence that RPI is informative about skill, not as a clean randomized causal estimate of how changing RPI mechanically changes fund performance.

4.5 Alternative explanations

- **Noise trading.** A low-RPI fund might trade randomly rather than privately informed. The performance results address this: random low-RPI trading should not produce abnormal returns.
- **Passive behavior.** A low-RPI fund might simply avoid trading. The paper controls for turnover and uses holdings-based performance measures.
- **Transaction costs.** High-RPI funds may trade more and lose to costs. The two-stage residualized RPI tests and turnover controls reduce this concern.
- **Analyst coverage.** Missing analyst data could select the sample. The paper checks analyst coverage and the number of analysts tracking portfolio stocks.
- **Information spillovers.** Analyst news about related stocks may matter. The paper constructs spillover-adjusted RPI using industry recommendation changes.

4.6 What the paper can and cannot distinguish

- It can distinguish funds whose trades are statistically explained by public analyst information from funds whose trades are not.
- It can show that low public-information reliance predicts higher performance and inflows.
- It cannot directly observe the private information held by managers.
- It cannot identify the exact content of the private information.
- It measures RPI only at discrete disclosure dates, not continuously.

Interpretation. RPI is a diagnostic proxy for skill, not a direct observation of information production.

5 Tables and figures

5.1 Table I: Summary of the data by RPI decile

Table I

Summary of the Data: Decile Portfolios

This table reports the variation in *TNA* (total net assets), dollar expenses, age, turnover, and total fund load across mutual fund deciles. We form deciles every quarter based on the *RPI* measure and average the variables for the deciles across all available quarters. *RPI* measures the reliance on public information and equals the unadjusted R^2 of the regression of percentage changes in fund managers' portfolio holdings on changes in analysts' past recommendations of up to four lags. The sample spans the period 1993Q1 to 2002Q4. The last row reports the correlation coefficients of each of the variables with *RPI*, along with their statistical significance.

Decile #	RPI (%)	TNA ($ Mil.)	Expenses ($ Mil.)	Age (Years.)	Turnover (%)	Load (%)
1	1.68	2325.3	27.21	13.29	54.14	1.44
2	4.45	1778.2	21.69	13.81	59.87	1.62
3	7.46	1461.0	17.97	14.22	70.32	1.73
4	10.94	960.1	11.91	12.86	72.68	1.69
5	15.23	772.0	9.88	13.33	76.30	1.71
6	20.54	716.9	9.25	13.01	80.55	1.69
7	27.09	615.9	8.13	12.95	84.41	1.65
8	36.05	509.0	6.82	13.35	90.39	1.57
9	49.92	407.3	5.54	12.88	97.12	1.68
10	76.14	321.7	4.57	11.83	110.85	1.52
	1.000	-0.1019^{***}	-0.0984^{***}	-0.0252^{***}	0.1890^{***}	-0.0189^{**}

***, **, * represent 1%, 5%, 10% confidence levels, respectively.

Read:

- RPI rises from 1.68% in decile 1 to 76.14% in decile 10.
- Fund size falls sharply with RPI: average TNA declines from about $2.3 billion to about $322 million.
- Turnover rises from 54.14% to 110.85%.
- Dollar expenses are lower among high-RPI funds, consistent with smaller funds and lower rents.

Economic message: High-RPI funds look smaller and more trading-intensive. The performance tests therefore need to control for size, expenses, and turnover.

5.2 Table II: RPI and performance

Table II

Relationship between *RPI* and Performance

This table reports the results of the regressions relating performance and *RPI*. In Panel A, we report the results of the regression $\alpha_{m,t} = \beta_0 + \beta_1 RPI_{m,t-1} + \gamma Controls_{m,t-1} + \epsilon_{m,t}$, where the dependent variable is the monthly factor-based measure $\alpha_{m,t}$. We use market-adjusted (CAPM) alpha, Fama and French's (1993) three-factor (market, size, value)-adjusted alpha, and Carhart's (1997) four-factor alpha, which adds momentum as a factor. We also include conditional measures of performance for each of the above unconditional measures using the approach of Ferson and Schadt (1996). In Panel B, we report the results of the panel regression $y_{m,t} = \beta_0 + \beta_1 RPI_{m,t-1} + \gamma Controls_{m,t-1} + \epsilon_{m,t}$, where the dependent variable $y_{m,t}$ is the monthly holding-based measure. Specifically, we include the Grinblatt–Titman measure (*GT*), the characteristic selectivity measure (*CS*), the characteristic timing measure (CT), and the average style measure (*AS*). *CS* is a measure of stock selection ability and is defined as $CS = \sum w_{j,t-1}[R_{j,t} - BR_t(j, t-1)]$, where $BR_t(j, t-1)$ is the period-t return of the benchmark portfolio to which stock j was allocated in period $t-1$ according to its size, value, and momentum characteristics. *CT* is a measure of style timing ability and is defined as $CT = \sum[w_{j,t-1}BR_t(j, t-1) - w_{j,t-5}BR_t(j, t-5)]$ and AS is a measure of style selection ability, and is defined as $AS = \sum[w_{j,t-5}BR_t(j, t-5)]$. *RPI* measures the reliance on public information and equals the unadjusted R^2 of the regression of percentage changes in fund managers' portfolio holdings on changes in analysts' past recommendations of up to four lags. Log*(TNA)* is the natural logarithm of total net assets lagged one quarter, Expenses denotes expenses lagged one year, *Log(Age)* is the natural logarithm of age lagged one quarter, Turnover is the turnover lagged one year, and *NMG* is the new money growth lagged one quarter. All regressions include quarterly time dummies. Standard errors reported in parentheses are corrected for heteroskedasticity and for the panel. Data are for the period 1993Q1 to 2002Q4.

Panel A: Factor-Based Measures (% per Month)

	Unconditional			Conditional		
	CAPM α	3-factor α	4-factor α	CAPM α	3-factor α	4-factor α
RPI_{t-1}	−0.23***	−0.09*	−0.17***	−0.37***	−0.25**	−0.35***
(%)	(0.06)	(0.05)	(0.06)	(0.12)	(0.11)	(0.10)
$Log(TNA)_{t-1}$	−2.79***	1.78**	−2.19**	−12.19***	−7.48***	−5.75***
	(0.87)	(0.83)	(0.86)	(1.61)	(1.50)	(1.44)
$Log\ (Age)_{t-1}$	0.01	−3.85*	1.06	24.55***	10.01***	12.33***
	(2.10)	(2.03)	(2.10)	(4.05)	(3.72)	(3.53)
$Expenses_{t-1}$	−9.55***	−1.92	−12.26***	−29.01***	−22.46***	−17.40***
(%)	(3.45)	(3.73)	(3.83)	(7.30)	(6.82)	(6.16)
$Turnover_{t-1}$	0.08***	0.14***	−0.07***	0.03	0.01	−0.09**
(%)	(0.02)	(0.02)	(0.02)	(0.04)	(0.04)	(0.04)
NMG_{t-1}	−0.01	0.24***	0.07	−0.32**	−0.48***	0.44***
	(0.09)	(0.08)	(0.09)	(0.16)	(0.15)	(0.14)
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	18,096	18,096	18,096	18,096	18,096	18,096

Table II—*Continued*

Panel B: Holding-Based Measures (% per Month)

	GT	CT	CS	AS
RPI_{t-1}	−0.16***	−0.02	−0.26***	−0.10***
(%)	(0.05)	(0.03)	(0.05)	(0.04)
$Log(TNA)_{t-1}$	−0.29	0.01	−2.88***	−1.71***
	(0.68)	(0.43)	(0.64)	(0.50)
$Log\ (Age)_{t-1}$	−0.21	−1.30	4.59***	1.16
	(1.65)	(1.11)	(1.57)	(1.22)
$Expenses_{t-1}$	4.47	1.34	−5.06*	−5.19**
(%)	(2.79)	(1.72)	(2.77)	(2.08)
$Turnover_{t-1}$	0.04**	0.05***	−0.03	−0.03**
(in %)	(0.02)	(0.01)	(0.02)	(0.01)
NMG_{t-1}	−0.01	−0.07**	0.11**	0.03
	(0.05)	(0.03)	(0.05)	(0.04)
Time fixed effects	Yes	Yes	Yes	Yes
Observations	18,964	18,964	18,964	18,964

***, **, * represent 1%, 5%, 10% confidence levels, respectively.

Read:

- In Panel A, RPI is negatively related to CAPM, three-factor, four-factor, and conditional alphas.
- The unconditional four-factor coefficient is −0.17 and statistically significant.
- A one-standard-deviation increase in RPI lowers four-factor alpha by about 0.46% per year.
- In Panel B, RPI is negatively related to GT, CS, and AS, but not CT.
- The CS coefficient is −0.26, indicating that the strongest holding-based evidence is stock selection.

Economic message: Funds that rely less on public analyst information perform better. The evidence points to stock-selection skill, not primarily timing skill.

Table III

Relationship between *RPI* and Fund Flows

This table reports the results of the regression $NetFlow_{m,t} = \beta_0 + \beta_1 RPI_{m,t-1} + \gamma Controls_{m,t-1} + \epsilon_{m,t}$. *NetFlow* measures the percentage flow of funds into mutual fund m between time $t-1$ and t. *RPI* measures the reliance on public information and equals the unadjusted-R^2 of the regression of percentage changes in fund managers' portfolio holdings on changes in analysts' past recommendations of up to four lags. R is the return on the fund portfolio lagged one quarter, Log(*TNA*) is the natural logarithm of total net assets lagged one quarter, Expenses denotes expenses lagged one year, Log(*Age*) is the natural logarithm of age lagged one quarter, Turnover is the turnover lagged one year, and other controls include the load of the fund lagged one quarter, raw returns of the fund in the last period, and the standard deviation of the fund returns based on the past 36 monthly returns. All regressions include quarterly time dummies. Standard errors reported in parentheses are corrected for heteroskedasticity and for the panel. Data are for the period 1993Q1 to 2002Q4.

		NMG_t	
RPI_{t-1}			-3.71^{***}
(%)			(0.51)
α^{4f}_{t-1}	0.73^{***}	0.084	0.086
	(0.06)	(0.08)	(0.07)
R_{t-1}		1.29^{***}	1.28^{***}
(%)		(0.08)	(0.08)
Log(TNA)_{t-1}	-6.02	-8.55	-16.50
	(9.60)	(9.90)	(9.88)
Log(Age)_{t-1}	-231.65^{***}	-231.27^{***}	-220.43^{***}
	(14.53)	(14.60)	(14.70)
Expenses_{t-1}	23.23	7.50	15.23
(%)	(29.67)	(29.70)	(29.60)
Turnover_{t-1}	0.61^{**}	0.49^{*}	0.68^{**}
(%)	(0.30)	(0.30)	(0.31)
Load_{t-1}	0.15^{***}	0.17^{***}	0.16^{***}
(%)	(0.06)	(0.05)	(0.05)
$\text{St. deviation}_{t-1}$	-25.63^{***}	-9.10	-5.80
	(7.46)	(7.10)	(7.50)
Time fixed effects	Yes	Yes	Yes
Observations	17,851	17,851	17,851

***, **, * represent 1%, 5%, 10% confidence levels, respectively.

5.3 Table III: RPI and fund flows

Read:

- Fund flows chase past performance, consistent with prior mutual fund literature.
- After controlling for past performance and other fund characteristics, the RPI coefficient is -3.71 and statistically significant.
- A one-standard-deviation increase in RPI implies roughly 3.71% lower annual flows.

Economic message: Investors appear to reward low-RPI funds even after conditioning on past returns. This suggests that RPI contains information about skill not fully summarized by realized performance.

5.4 Table IV: Alternative RPI based on coefficient significance

Read:

- The alternative RPI^{β} measure is based on the absolute t-statistics of the lagged public-information coefficients.
- The negative relationship with factor-adjusted and holding-based performance remains.
- The paper reports that the flow result also remains: the RPI coefficient in the flow regression is -3.88 and significant at the 1% level.

Economic message: The core result is not mechanically tied to defining RPI as an R^2.

Table IV

Relationship between RPI^{β} and Performance

This table reports the results of the regressions relating performance and modified *RPI* (RPI^{β}). In Panel A, we report the results of the regression $\alpha_{m,t} = \beta_0 + \beta_1 RPI^{\beta}_{m,t-1} + \gamma Controls_{m,t-1} + \epsilon_{m,t}$, where the dependent variable is the factor-based measure, $\alpha_{m,t}$. The risk-adjusted measures include the monthly unconditional (columns two-four) and conditional (columns five-seven) CAPM, three-factor, and four-factor alpha. The conditional measures are derived using the modified procedure of Ferson and Schadt (1996), proposed by Wermers (2000). In Panel B, we report the results of the panel regression $y_{m,t} = a + \beta_1 RPI^{\beta}_{m,t-1} + \gamma Controls_{m,t-1} + \epsilon_{m,t}$, where the dependent variable $y_{m,t}$ is the monthly holding-based measure. We use respectively the Grinblatt and Titman measure (*GT*), the characteristic selectivity measure (*CS*), the characteristic timing measure (*CT*), and the average style measure (*AS*) of Daniel et al. (1997). *CS* is a measure of stock selection ability and is defined as $CS = \sum w_{j,t-1}[R_{j,t} - BR_t(j,t-1)]$ where $BR_t(j,t-1)$ is the period-t return of the benchmark portfolio to which stock j was allocated in period $t-1$ according to its size, value, and momentum characteristics. *CT* is a measure of the style-timing ability and is defined as $CT = \sum[w_{j,t-1}BR_t(j,t-1) - w_{j,t-5}BR_t(j,t-5)]$ and *AS* is a measure of the style-selection ability and is defined as $AS = \sum[w_{j,t-5}BR_t(j,t-5)]$. RPI^{β} is an alternative specification to measure the reliance on public information and is calculated as per equation (14). Log*(TNA)* is the natural logarithm of total net assets lagged one quarter, Expenses denotes expenses lagged 1 year, Log*(Age)* is the natural logarithm of age lagged one quarter, Turnover is the turnover lagged 1 year, and *NMG* is the new money growth lagged one quarter. All regressions include quarterly time dummies. Standard errors reported in parentheses are corrected for heteroskedasticity and for the panel. Data are for the period 1993Q1 to 2002Q4.

Panel A: Factor-Based Measures (% per Month)

	Unconditional			Conditional		
	CAPM α	3-factor α	4-factor α	CAPM α	3-factor α	4-factor α
RPI^{β}_{t-1}	−0.25**	−0.12**	−0.19**	−0.42***	−0.29**	−0.38**
(in %)	(0.12)	(0.05)	(0.07)	(0.16)	(0.14)	(0.17)
$Log(TNA)_{t-1}$	−2.71***	1.88***	−2.01***	−12.00***	−7.40***	−6.01***
	(0.66)	(0.63)	(0.72)	(1.41)	(1.53)	(1.31)
$Log\ (Age)_{t-1}$	0.03	−3.99*	1.01	25.40**	10.06***	12.89***
	(2.21)	(2.00)	(2.34)	(4.20)	(3.03)	(3.05)
$Expenses_{t-1}$	−9.51***	−1.90	−12.21***	−29.10***	−22.64***	−17.45***
(in %)	(3.50)	(3.32)	(3.80)	(7.23)	(6.23)	(6.45)
$Turnover_{t-1}$	0.08***	0.14***	−0.07***	0.03	0.01	−0.11***
(in %)	(0.01)	(0.02)	(0.02)	(0.05)	(0.05)	(0.03)
NMG_{t-1}	−0.04	0.24***	0.07	−0.32***	−0.48***	0.44***
	(0.09)	(0.08)	(0.10)	(0.10)	(0.13)	(0.16)
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	18,096	18,096	18,096	18,096	18,096	18,096

(*continued*)

Table IV—*Continued*

Panel B: Holding-Based Measures (% per Month)

	GT	CT	CS	AS
RPI^{β}_{t-1}	−0.19***	−0.04	−0.31**	−0.13***
(%)	(0.06)	(0.03)	(0.15)	(0.04)
$Log(TNA)_{t-1}$	−0.20	0.07	−2.97***	−1.88***
	(0.99)	(0.48)	(0.51)	(0.53)
$Log\ (Age)_{t-1}$	−0.12	−1.41	5.04***	1.11
	(1.80)	(2.01)	(1.60)	(1.45)
$Expenses_{t-1}$	4.71	1.45	−5.66*	−5.97**
(%)	(3.01)	(1.88)	(2.93)	(1.99)
$Turnover_{t-1}$	0.04***	0.04**	−0.06*	−0.02**
(%)	(0.01)	(0.02)	(0.04)	(0.01)
NMG_{t-1}	−0.07	−0.08**	0.13**	0.02
	(0.06)	(0.04)	(0.05)	(0.04)
Time fixed effects	Yes	Yes	Yes	Yes
Observations	18,964	18,964	18,964	18,964

***, **, * represent 1%, 5%, 10% confidence levels respectively.

5.5 Table V: Alternative public information based on earnings forecasts

Table V

Relationship between RPI^{earn} and Performance

This table reports the results of the regressions relating performance and modified *RPI* (RPI^{earn}). In Panel A, we report the results of the regression $\alpha_{m,t} = \beta_0 + \beta_1 RPI^{earn}{}_{m,t-1} + \gamma Controls_{m,t-1} + \epsilon_{m,t}$, where the dependent variable is the factor-based measure, $\alpha_{m,t}$. The risk-adjusted measures include the monthly unconditional (columns two-four) and conditional (columns five-seven) CAPM, three-factor, and four-factor alpha. The conditional measures are derived using the modified procedure of Ferson and Schadt (1996), proposed by Wermers (2000). In Panel B, we report the results of the panel regression $y_{m,t} = a + \beta_1 RPI^{earn}{}_{m,t-1} + \gamma Controls_{m,t-1} + \epsilon_{m,t}$, where the dependent variable $y_{m,t}$ is the monthly holding-based measure. We use respectively the Grinblatt and Titman measure (*GT*), the characteristic selectivity measure (*CS*), the characteristic timing measure (*CT*), and the average style measure (*AS*) of Daniel et al. (1997). *CS* is a measure of stock selection ability and is defined as $CS = \sum w_{j,t-1}[R_{j,t} - BR_t(j,t-1)]$ where $BR_t(j,t-1)$ is the period-t return of the benchmark portfolio to which stock j was allocated in period $t-1$ according to its size, value, and momentum characteristics. *CT* is a measure of the style-timing ability and is defined as $CT = \sum[w_{j,t-1}BR_t(j,t-1) - w_{j,t-5}BR_t(j,t-5)]$ and *AS* is a measure of the style-selection ability and is defined as $AS = \sum[w_{j,t-5}BR_t(j,t-5)]$. RPI^{earn} is an alternative measure of reliance on public information and equals the unadjusted R^2 of the regression of changes in fund managers' portfolio holdings on changes in analysts' past earning forecasts of up to four lags. Log*(TNA)* is the natural logarithm of total net assets lagged one quarter, Expenses denotes expenses lagged 1 year, Log*(Age)* is the natural logarithm of age lagged one quarter, Turnover is the turnover lagged 1 year, and *NMG* is the new money growth lagged one quarter. All regressions include quarterly time dummies. Standard errors reported in parentheses are corrected for heteroskedasticity and for the panel. Data are for the period 1993Q1 to 2002Q4.

Panel A: Factor-Based Measures (% per Month)

	Unconditional			Conditional		
	CAPM α	3-factor α	4-factor α	CAPM α	3-factor α	4-factor α
RPI^{earn}_{t-1}	−0.30*	−0.17*	−0.26**	−0.44**	−0.36*	−0.39**
(%)	(0.17)	(0.09)	(0.12)	(0.22)	(0.21)	(0.20)
$Log(TNA)_{t-1}$	−2.89***	1.89**	−2.91**	−11.12***	−8.99***	−5.52***
	(0.80)	(0.69)	(0.90)	(2.03)	(3.22)	(2.03)
$Log\ (Age)_{t-1}$	0.02	−5.38**	2.61	20.15***	12.11***	12.07***
	(2.19)	(2.20)	(2.89)	(5.11)	(3.04)	(3.05)
$Expenses_{t-1}$	−10.01***	−1.97	−12.60***	−31.92***	−24.62***	−14.70***
(%)	(3.51)	(3.77)	(3.38)	(8.83)	(8.42)	(3.61)
$Turnover_{t-1}$	0.09***	0.12***	−0.09***	0.02	0.02	−0.07**
(%)	(0.03)	(0.03)	(0.03)	(0.05)	(0.05)	(0.03)
NMG_{t-1}	−0.10	0.12***	0.71	−0.12**	−0.18***	0.14***
	(0.09)	(0.03)	(0.59)	(0.06)	(0.05)	(0.05)
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	18,096	18,096	18,096	18,096	18,096	18,096

(*continued*)

Table V—*Continued*

Panel B: Holding-Based Measures (% per Month)

	GT	CT	CS	AS
RPI^{earn}_{t-1}	−0.29*	−0.15	−0.48**	−0.21**
(%)	(0.16)	(0.10)	(0.20)	(0.11)
$Log(TNA)_{t-1}$	−0.19	0.11	−2.20***	−1.91***
	(0.80)	(0.31)	(0.49)	(0.56)
$Log\ (Age)_{t-1}$	−0.71	−0.90	4.01***	1.68
	(1.55)	(0.99)	(0.60)	(1.18)
$Expenses_{t-1}$	3.71	1.43	−5.62*	−5.91**
(%)	(3.91)	(2.27)	(2.89)	(2.01)
$Turnover_{t-1}$	0.03**	0.04***	−0.07	−0.04**
(%)	(0.01)	(0.01)	(0.06)	(0.02)
NMG_{t-1}	−0.22	−0.13**	0.16**	0.13
	(0.15)	(0.05)	(0.07)	(0.14)
Time fixed effects	Yes	Yes	Yes	Yes
Observations	18,964	18,964	18,964	18,964

***, **, * represent 1%, 5%, 10% confidence levels, respectively.

Read:

- RPI^{earn} replaces analyst recommendations with changes in analysts' earnings forecasts.

- RPI^{earn} has similar cross-sectional variation: mean 33.4%, median 27.1%, standard deviation 20.3%.
- Low-RPI^{earn} funds outperform using factor and holding-based measures.
- The flow regression remains negative and significant, with coefficient -2.72.

Economic message: The results generalize beyond one specific public-information proxy.

5.6 Table VI: Managerial turnover and manager fixed effects

Read:

- The paper identifies 2,193 managers and 952 manager changes.
- Adding manager fixed effects increases adjusted R^2 for RPI from 0.27 to 0.35.
- Manager fixed effects are jointly significant.
- Similar patterns hold for RPI^{earn}.
- The evidence suggests RPI tends to fall after managerial turnover, especially after poor prior performance.

Economic message: RPI is not only a fund-level attribute. Manager-specific characteristics explain meaningful variation in reliance on public information.

Table VI

Relationship between *RPI* (*RPI^{earn}*) and Managerial Turnover

This table reports the results of the regressions relating manager turnover and *RPI* (*RPI^{earn}*). The nature of our identification strategy closely follows that of Bertrand and Schoar (2003), who examine whether corporate policies are affected by managerial styles. Specifically, we first take a benchmark specification, from which we derive the residual *RPI*. This benchmark specification is estimated at the fund-year level after controlling for any average differences across funds and years, as well as for any fund-year specific shocks, such as flows, that might affect the *RPI* of a fund. We then ask how much of the variance in the residual *RPI* can be attributed to manager-specific effects. In Panel A, we estimate the following regression: $RPI_{it} = \alpha_t + \gamma_i + \beta X_{it} + \lambda_{mgr}$, where α_t are year fixed effects, γ_i are fund fixed effects, X_{it} represents a vector of time-varying fund level controls. λ_{mgr} in equation (15) denotes manager fixed effects. In Panel B, we replace *RPI* by *RPI^{earn}*. In each row in both panels, we report the F-test statistics and adjusted R^2 from the estimation of equations with relevant fund and manager controls. More concretely, in the first row of the table we report the fit of a benchmark specification that includes only fund fixed effects and year fixed effects. In the second row we also include time-varying fund controls, while in the third row we add manager fixed effects. The second and the third rows, respectively, also report the adjusted R^2 after adding time-varying fund controls and manager fixed effects. The third row also reports F-statistics from tests of the joint significance of the manager fixed effects. Finally, the fourth row also includes the interaction of manager fixed effects with Bad, a dummy variable that takes a value of one in time period t if the market-adjusted abnormal return of the fund in the past three consecutive quarters is negative and zero otherwise. *RPI^{earn}* is an alternative measure of reliance on public information and equals the unadjusted R^2 of the regression of changes in managers' portfolio holdings on changes in analysts' past earning forecasts of up to four lags. *RPI* measures the reliance on public information and equals the unadjusted R^2 of the regression of percentage changes in managers' portfolio holdings on changes in analysts' past recommendations of up to four lags. Fund controls include Log*(TNA)*, the natural logarithm of total net assets lagged one quarter; Expenses, expenses lagged 1 year; Log*(Age)*, the natural logarithm of age lagged one quarter; Turnover, the turnover lagged 1 year; and *NMG*, the new money growth lagged one quarter. Data are for the period 1993Q1 to 2002Q4.

	Fund Controls	Fund and Time Fixed Effects	Manager Fixed Effects	Manager Fixed Effects × Bad	N (Observations)	Adjusted R^2
Panel A: F-tests on Manager Fixed Effects Using *RPI*						
RPI	No	Yes			18,096	0.25
RPI	Yes	Yes			18,096	0.27
RPI	Yes	Yes	18.77 (<0.0001, 943)		18,096	0.35
RPI	Yes	Yes	1.21 (0.5327, 943)	5.93 (<0.0001, 688)	18,096	0.36
Panel B: F-Tests on Manager Fixed Effects Using *RPI^{earn}*						
RPI	No	Yes			18,096	0.19
RPI	Yes	Yes			18,096	0.22
RPI	Yes	Yes	15.44 (<0.0001, 943)		18,096	0.27
RPI	Yes	Yes	0.92 (0.3789, 943)	6.59 (<0.0001, 688)	18,096	0.30

Table VII

***RPI^s* and Performance**

This table reports the results of the regressions relating performance and modified *RPI* (*RPI^s*). In Panel A, we report the results of the regression $\alpha_{m,t} = \beta_0 + \beta_1 RPI^s{}_{m,t-1} + \gamma Controls_{m,t-1} + \epsilon_{m,t}$, where the dependent variable is the factor-based measure, $\alpha_{m,t}$. The risk-adjusted measures include the monthly unconditional (columns two-four) and conditional (columns five-seven) CAPM, three-factor, and four-factor alpha. The conditional measures are derived using the modified procedure of Ferson and Schadt (1996), proposed by Wermers (2000). In Panel B, we report the results of the panel regression $y_{m,t} = a + \beta_1 RPI^s{}_{m,t-1} + \gamma Controls_{m,t-1} + \epsilon_{m,t}$, where the dependent variable $y_{m,t}$ is the monthly holding-based measure. We use respectively the Grinblatt and Titman measure (*GT*), the characteristic selectivity measure (*CS*), the characteristic timing measure (*CT*), and the average style measure (*AS*) of Daniel et al. (1997). *CS* is a measure of stock selection ability and is defined as $CS = \sum w_{j,t-1}[R_{j,t} - BR_t(j,t-1)]$ where $BR_t(j,t-1)$ is the period-t return of the benchmark portfolio to which stock j was allocated in period $t-1$ according to its size, value, and momentum characteristics. *CT* is a measure of the style-timing ability and is defined as $CT = \sum[w_{j,t-1}BR_t(j,t-1) - w_{j,t-5}BR_t(j,t-5)]$ and *AS* is a measure of the style-selection ability and is defined as $AS = \sum[w_{j,t-5}BR_t(j,t-5)]$. *RPI^s* measures the reliance on public information and equals the unadjusted R^2 of the regression of changes in managers' portfolio holdings on explanatory variables given in equation (24). Log*(TNA)* is the natural logarithm of total net assets lagged one quarter, Expenses denotes expenses lagged 1 year, Log*(Age)* is the natural logarithm of age lagged one quarter, Turnover is the turnover lagged 1 year, and *NMG* is the new money growth lagged one quarter. All regressions include quarterly time dummies. Standard errors reported in parentheses are corrected for heteroskedasticity and for the panel. Data are for the period 1993Q1 to 2002Q4.

Panel A: Factor-Based Measures (% per Month)						
	Unconditional			Conditional		
	CAPM α	3-factor α	4-factor α	CAPM α	3-factor α	4-factor α
RPI^s_{t-1} (%)	-0.20^{***} (0.12)	-0.14^{**} (0.07)	-0.18^{**} (0.08)	-0.38^{***} (0.14)	-0.26^{**} (0.17)	-0.32^{**} (0.18)
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	17,944	17,944	17,944	17,944	17,944	17,944

Panel B: Holding-Based Measures (% per Month)				
	GT	CT	CS	AS
RPI^s_{t-1} (%)	-0.16^{***} (0.04)	-0.03 (0.04)	-0.35^{**} (0.19)	-0.14^{***} (0.04)
Time fixed effects	Yes	Yes	Yes	Yes
Observations	18,820	18,820	18,820	18,820

***, **, * represent 1%, 5%, 10% confidence levels, respectively. We do not report estimates on fund

5.7 Table VII: Spillover-adjusted RPI

Table VII

***RPI*^{s} and Performance**

This table reports the results of the regressions relating performance and modified *RPI* (RPI^s). In Panel A, we report the results of the regression $\alpha_{m,t} = \beta_0 + \beta_1 RPI^s{}_{m,t-1} + \gamma Controls_{m,t-1} + \epsilon_{m,t}$, where the dependent variable is the factor-based measure, $\alpha_{m,t}$. The risk-adjusted measures include the monthly unconditional (columns two-four) and conditional (columns five-seven) CAPM, three-factor, and four-factor alpha. The conditional measures are derived using the modified procedure of Ferson and Schadt (1996), proposed by Wermers (2000). In Panel B, we report the results of the panel regression $y_{m,t} = a + \beta_1 RPI^s{}_{m,t-1} + \gamma Controls_{m,t-1} + \epsilon_{m,t}$, where the dependent variable $y_{m,t}$ is the monthly holding-based measure. We use respectively the Grinblatt and Titman measure (*GT*), the characteristic selectivity measure (*CS*), the characteristic timing measure (*CT*), and the average style measure (*AS*) of Daniel et al. (1997). *CS* is a measure of stock selection ability and is defined as $CS = \sum w_{j,t-1}[R_{j,t} - BR_t(j,t-1)]$ where $BR_t(j,t-1)$ is the period-t return of the benchmark portfolio to which stock j was allocated in period $t-1$ according to its size, value, and momentum characteristics. *CT* is a measure of the style-timing ability and is defined as $CT = \sum[w_{j,t-1}BR_t(j,t-1) - w_{j,t-5}BR_t(j,t-5)]$ and *AS* is a measure of the style-selection ability and is defined as $AS = \sum[w_{j,t-5}BR_t(j,t-5)]$. RPI^s measures the reliance on public information and equals the unadjusted R^2 of the regression of changes in managers' portfolio holdings on explanatory variables given in equation (24). Log*(TNA)* is the natural logarithm of total net assets lagged one quarter, Expenses denotes expenses lagged 1 year, Log*(Age)* is the natural logarithm of age lagged one quarter, Turnover is the turnover lagged 1 year, and *NMG* is the new money growth lagged one quarter. All regressions include quarterly time dummies. Standard errors reported in parentheses are corrected for heteroskedasticity and for the panel. Data are for the period 1993Q1 to 2002Q4.

Panel A: Factor-Based Measures (% per Month)

	Unconditional			Conditional		
	CAPM α	3-factor α	4-factor α	CAPM α	3-factor α	4-factor α
RPI^s_{t-1}	-0.20^{***}	-0.14^{**}	-0.18^{**}	-0.38^{***}	-0.26^{**}	-0.32^{**}
(%)	(0.12)	(0.07)	(0.08)	(0.14)	(0.17)	(0.18)
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	17,944	17,944	17,944	17,944	17,944	17,944

Panel B: Holding-Based Measures (% per Month)

	GT	CT	CS	AS
RPI^s_{t-1}	-0.16^{***}	-0.03	-0.35^{**}	-0.14^{***}
(%)	(0.04)	(0.04)	(0.19)	(0.04)
Time fixed effects	Yes	Yes	Yes	Yes
Observations	18,820	18,820	18,820	18,820

***, **, * represent 1%, 5%, 10% confidence levels, respectively. We do not report estimates on fund

Read:

- RPI^s includes both direct recommendation changes and recommendation changes for other stocks in the same three-digit SIC industry.
- RPI^s remains negatively related to factor alphas and holding-based measures.
- The paper reports that the fund-flow coefficient is also negative, -2.93, and significant at the 1% level.

Economic message: The result is not driven by ignoring information spillovers across related stocks in the portfolio.

5.8 Table VIII: Two-stage residualized RPI

Read:

- In the first stage, RPI is residualized on fund characteristics, including turnover.
- In the second stage, residual RPI predicts factor-based performance.
- The residual coefficient is negative in both panel and Fama–MacBeth-style cross-sectional specifications.

Economic message: The RPI–performance relationship is not simply a turnover or fund-characteristics story.

Table VIII

Relationship between *RPI* and Performance: Two-Stage Analysis

This table reports the results of the panel and cross-sectional regression relating performance and residuals from the panel and cross-sectional regression of *RPI* on other fund characteristics (first stage). The results of the first stage are unreported for brevity. In columns two-four, we report the results of the panel regression (second stage; first stage estimated in panel): $\alpha_{m,t} = \beta_0 + \beta_1 Residual_{m,t-1} + \epsilon_{m,t}$, where the dependent variable is the monthly factor-based measure $\alpha_{m,t}$. We use market-adjusted (CAPM) alpha, three-factor (market, size, value) adjusted alpha of Fama and French (1993), and four-factor alpha of Carhart (1997), which additionally includes momentum as a factor. In columns five–seven, we report the results of the cross sectional regression (alternative specification for second stage; first stage estimated as cross-section) with standard errors calculated as in Fama and MacBeth (1973). *RPI* measures the reliance on public information and equals the unadjusted R^2 of the regression of percentage changes in managers' portfolio holdings on changes in analysts' past recommendations of up to four lags. Fund characteristics include Log*(TNA)* is the natural logarithm of total net assets lagged one quarter, Expenses denotes expenses lagged 1 year, Log*(Age)* is the natural logarithm of age lagged one quarter, Turnover is the turnover lagged 1 year, and *NMG* is the new money growth lagged one quarter. Data are for the period 1993Q1 to 2002Q4.

	Factor-Based Measures (% per Month)					
	Panel Regression			Cross-Sectional Regression		
	CAPM α	3-factor α	4-factor α	CAPM α	3-factor α	4-factor α
Residual	-0.29^{***}	-0.12^{**}	-0.22^{***}	-0.19^{*}	-0.11^{*}	-0.16^{***}
(%)	(0.06)	(0.06)	(0.06)	(0.10)	(0.06)	(0.07)
Time fixed effects	Yes	Yes	Yes			
Observations	18,096	18,096	18,096	33	33	33

***, **, * represent 1%, 5%, 10% confidence levels, respectively.

Table IX

Relationship between *RPI* and Risk-Taking

This table reports results related to managerial risk taking and *RPI*. Specifically, we report the results of the regression of total and unsystematic risk of a fund on its *RPI*, lagged returns, *TNA*, Age, Expenses, and Turnover. *TotRisk* measures the total fund risk and is calculated as a standard deviation of a fund's past 36 monthly returns. *UnsysRisk* measures the level of fund unsystematic risk and is calculated as a residual of the regression of fund returns on four risk/style factors of Carhart (1997). *RPI* measures the reliance on public information and equals the unadjusted R^2 of the regression of percentage changes in managers' portfolio holdings on changes in analysts' past recommendations of up to four lags. *R* is the return on the fund portfolio lagged one quarter, Log*(TNA)* is the natural logarithm of total net assets lagged one quarter, Expenses denotes expenses lagged 1 year, Log*(Age)* is the natural logarithm of age lagged one quarter, Turnover is the turnover lagged 1 year. All regressions include quarterly time dummies. Standard errors reported in parentheses are corrected for heteroskedasticity and for the panel. Data are for the period 1993Q1 to 2002Q4.

	TotRisk_t	UnsysRisk_t
RPI_{t-1}	0.54^{***}	0.41^{***}
(%)	(0.06)	(0.03)
R_{t-1}	-13.14^{***}	-1.37^{**}
(%)	(1.12)	(0.65)
Log(TNA)_{t-1}	10.31^{***}	0.79^{*}
	(0.83)	(0.47)
Log(Age)_{t-1}	-22.67^{***}	-5.58^{***}
	(2.00)	(1.13)
Expenses_{t-1}	65.27^{***}	44.30^{***}
(%)	(4.74)	(3.29)
Turnover_{t-1}	0.70^{***}	0.31^{***}
(in %)	(0.04)	(0.02)
Time fixed effects	Yes	Yes
Observations	18,131	18,131

***, **, * represent 1%, 5%, 10% confidence levels, respectively.

5.9 Table IX: RPI and risk-taking

Read:

- High-RPI funds take more total risk: coefficient 0.54.
- High-RPI funds also take more unsystematic risk: coefficient 0.41.
- Both effects are statistically significant.

Economic message: Managers with higher reliance on public information appear to take more risk, consistent with lower-skill managers gambling for better relative performance.

5.10 Table X: RPI and trade direction

Table X

Relationship between *RPI* and Trade Direction

This table reports the average values of the coefficients' estimates of the regression $\%\Delta Hold_{i,t} = \alpha + \beta_1 \Delta R_{i,t-1} + \beta_2 \Delta R_{i,t-2} + \beta_3 \Delta R_{i,t-3} + \beta_4 \Delta R_{i,t-4} + \epsilon$, where the dependent variable $\%\Delta Hold_{i,t}$ is the percentage change in the stock split-adjusted holdings of stock i from time $t-1$ to t. $R_{i,t-p}$ is the change in recommendation of the consensus forecast of stock i from time $t-p-1$ to time $t-p$, where $p = 1, 2, 3, 4$ is the number of the p^{th} forecast, with $R = 0$ if the forecast did not change during the two consecutive forecasts. The loadings are obtained using a two-stage procedure. In the first stage, for each time period we run the holding regression. In the second stage, we take the time-series average of the resulting estimates within each decile portfolio sorted with respect to *RPI* in each time period. Standard errors, calculated as in Fama and MacBeth (1973), are controlled for heteroskedasticity using the Newey and West (1987) procedure. The data span the period 1993Q1 to 2002Q4.

RPI Decile	*RPI* (%)	β_1	β_2	β_3	β_4
1	1.34***	0.76	−6.86	3.68	4.85
	(0.04)	(7.80)	(7.05)	(9.32)	(6.06)
2	4.38***	3.60	6.02	−11.18	−1.12
	(0.14)	(5.22)	(6.44)	(7.98)	(5.74)
3	7.91***	3.20	−3.20	6.57	1.02
	(0.27)	(0.04)	(9.14)	(9.02)	(4.56)
4	12.30***	−10.81*	−19.92***	16.08	4.52
	(0.43)	(6.59)	(8.20)	(9.99)	(5.52)
5	17.54***	−11.44*	−12.17**	−10.22	−6.32
	(0.64)	(6.75)	(6.13)	(7.01)	(4.93)
6	23.88***	−17.08**	−26.90***	−10.77	−4.68
	(0.85)	(7.40)	(7.81)	(8.62)	(4.05)
7	32.10***	−38.97***	−31.44***	−8.50	0.78
	(1.11)	(6.82)	(9.25)	(7.20)	(5.81)
8	42.54***	−37.93***	−45.42***	−10.70	−3.88
	(1.30)	(12.10)	(10.09)	(10.57)	(7.40)
9	56.92***	−62.42***	−27.17*	3.59	−6.63
	(1.43)	(14.49)	(16.04)	(11.73)	(6.70)
10	81.19***	−73.17***	−10.40	3.80	−9.41
	(1.17)	(17.90)	(26.55)	(27.35)	(8.79)

***, **, * represent 1%, 5%, 10% confidence levels, respectively.

Read:

- Low-RPI funds do not show a strong systematic direction relative to public recommendation changes.
- High-RPI funds tend to trade against recommendation changes in several lag specifications.

Economic message: RPI measures how much public information explains trading, not whether the fund mechanically follows or opposes analyst recommendations.

6 Short summary

- The paper asks whether fund managers' reliance on public information reveals managerial skill.
- The theoretical model predicts that more skilled managers, with more precise private information, should respond less to public signals.
- The empirical measure RPI is the R^2 from a holdings-change regression on lagged analyst recommendation changes.
- Low-RPI funds outperform high-RPI funds using factor-adjusted returns and holding-based measures.
- The holding-based evidence is strongest for stock selection and style selection, not timing.
- Fund flows are negatively related to RPI after controlling for past performance, implying that investors appear to value the information in RPI.

- Robustness tests show that the results survive alternative RPI definitions, alternative public-information sets, manager fixed effects, spillover controls, turnover controls, and fund size/style checks.
- The paper's main contribution is to turn the idea of "using private information" into an observable holdings-based measure.

7 Conclusion

7.1 Core conclusion

Kacperczyk and Seru show that a fund manager's reliance on public information is informative about skill. Managers whose trades are less explained by public analyst-recommendation changes tend to deliver better subsequent performance and attract more subsequent flows.

7.2 Economic intuition

A skilled manager has information or judgment that is not already fully captured by public signals. Therefore, the skilled manager does not need to adjust holdings strongly when public analyst information changes. Less-skilled managers, by contrast, rely more heavily on public signals and may also take more risk to improve their records.

7.3 Takeaway

The paper reframes mutual fund skill evaluation. Instead of asking only whether a manager earned high returns, it asks whether the manager's trades look like they came from information that everyone already had. Low reliance on public information becomes a useful complement to traditional performance measures.

7.4 Limitations

- RPI does not reveal the content of the manager's private information.
- RPI is data-intensive because it requires holdings and public-information data.
- RPI is measured only at discrete portfolio disclosure dates.
- The design provides strong predictive and interpretive evidence, but it is not a randomized causal experiment.